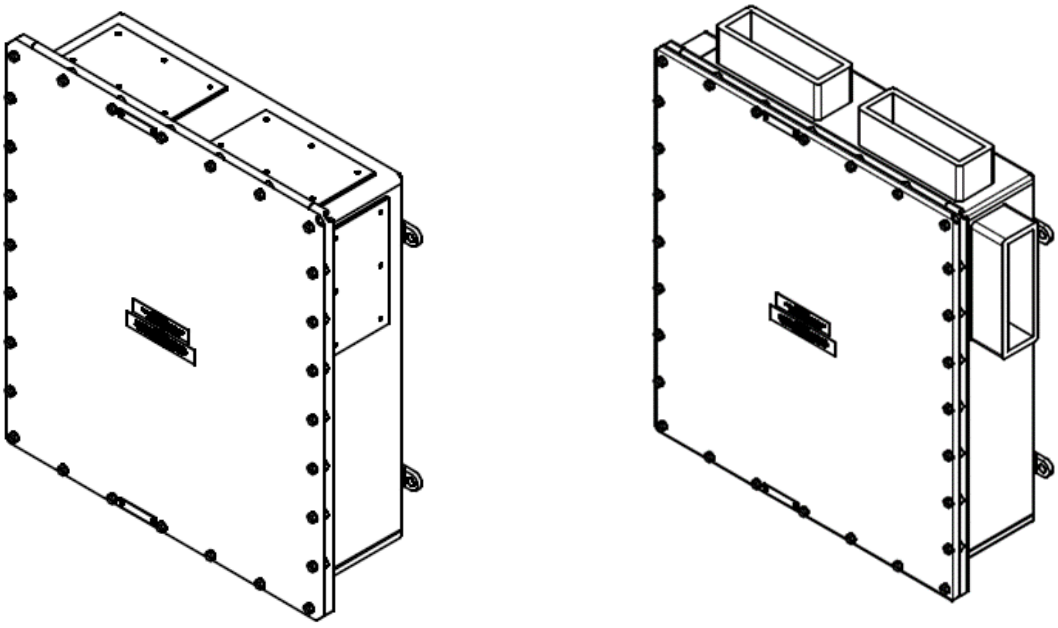
DETAIL SPECIFICATION SHEET

INTERCONNECTION BOX, CABLE ROUTING, FIBER OPTIC, SPRAYTIGHT, SUBMERSIBLE, 711 X 787 MM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24728.



NOTES:

1. For identification of subcomponent parts (circled numbers), see table I.
2. Dimensions are in millimeters.
3. The tolerances shall be 0.XX = 0.381 millimeter, 0.XXX = 0.127 millimeter, angles = 0°30' (0.XX = 0.015 inch, 0.XXX = 0.005 inch, angles = 0°30').

FIGURE 1. Spraytight (M24728/12-01, left) and submersible (M24728/12-02, right) fiber interconnection boxes.

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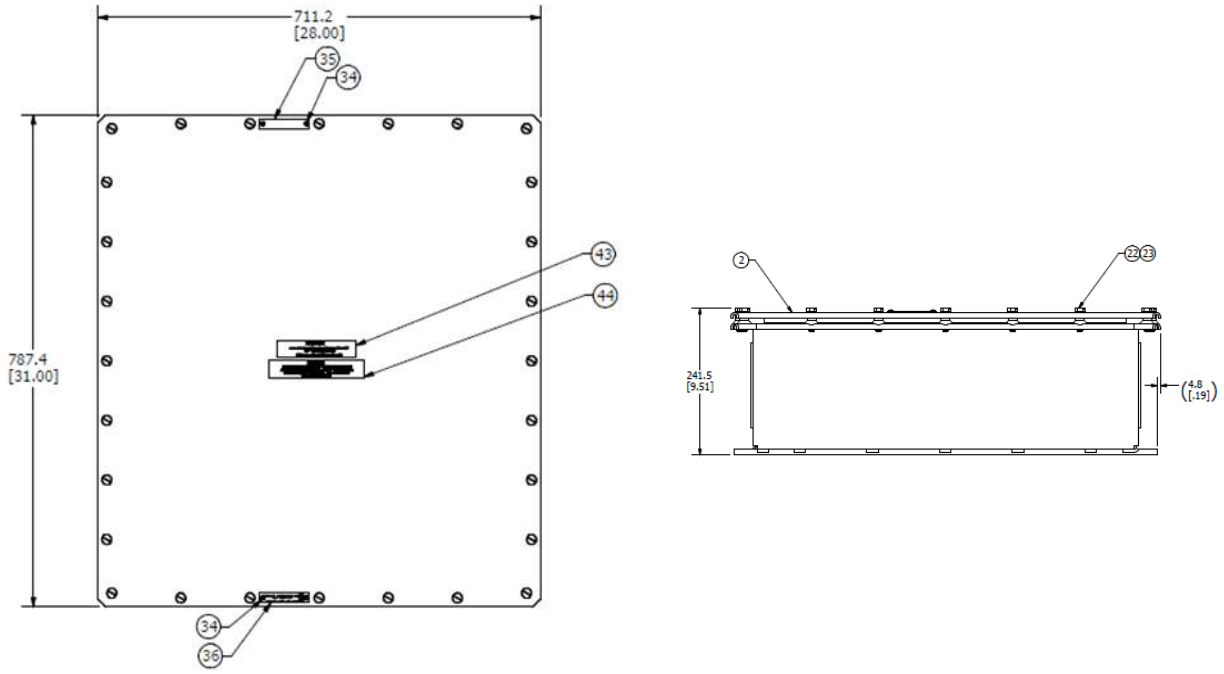


FIGURE 2. Enclosure assembly for M24728/12-01.

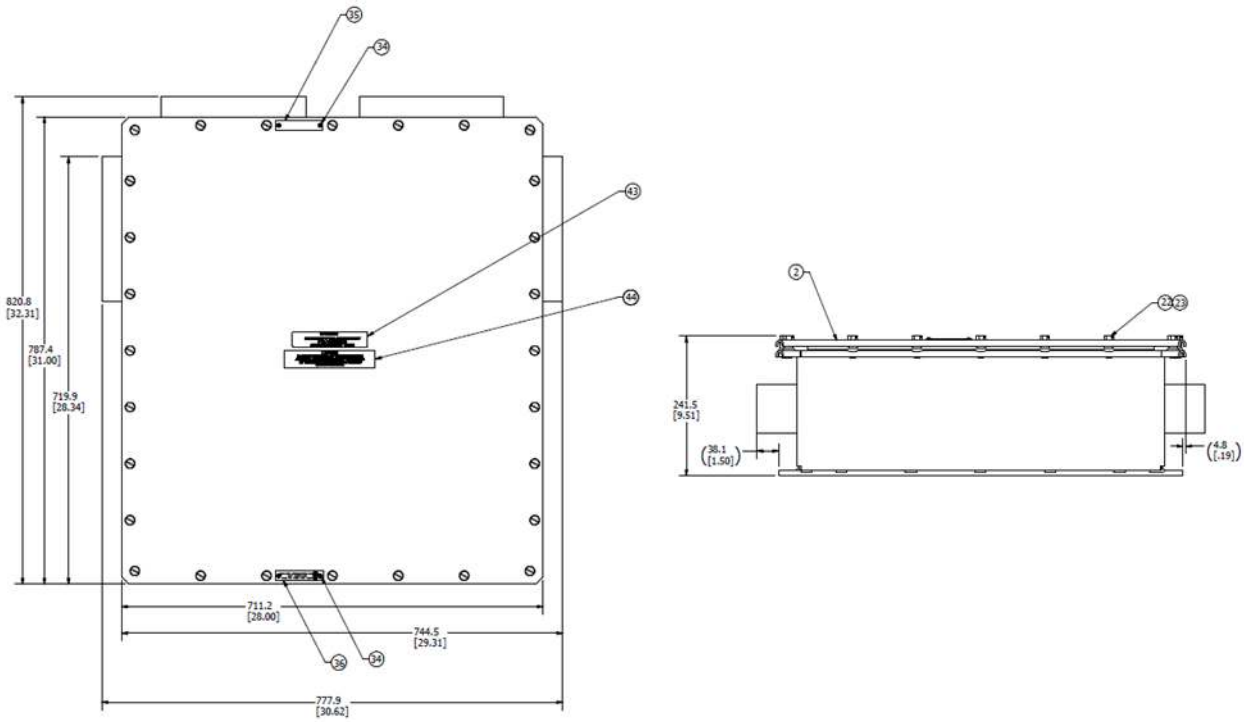
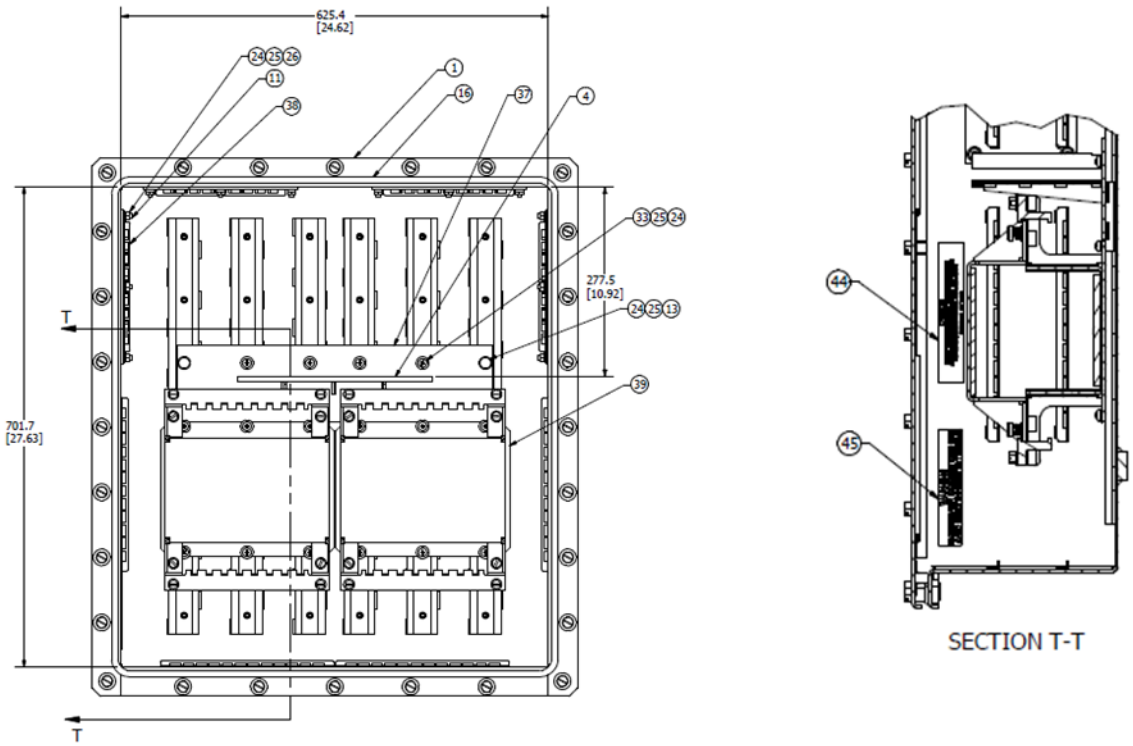


FIGURE 3. Enclosure assembly for M24728/12-02.

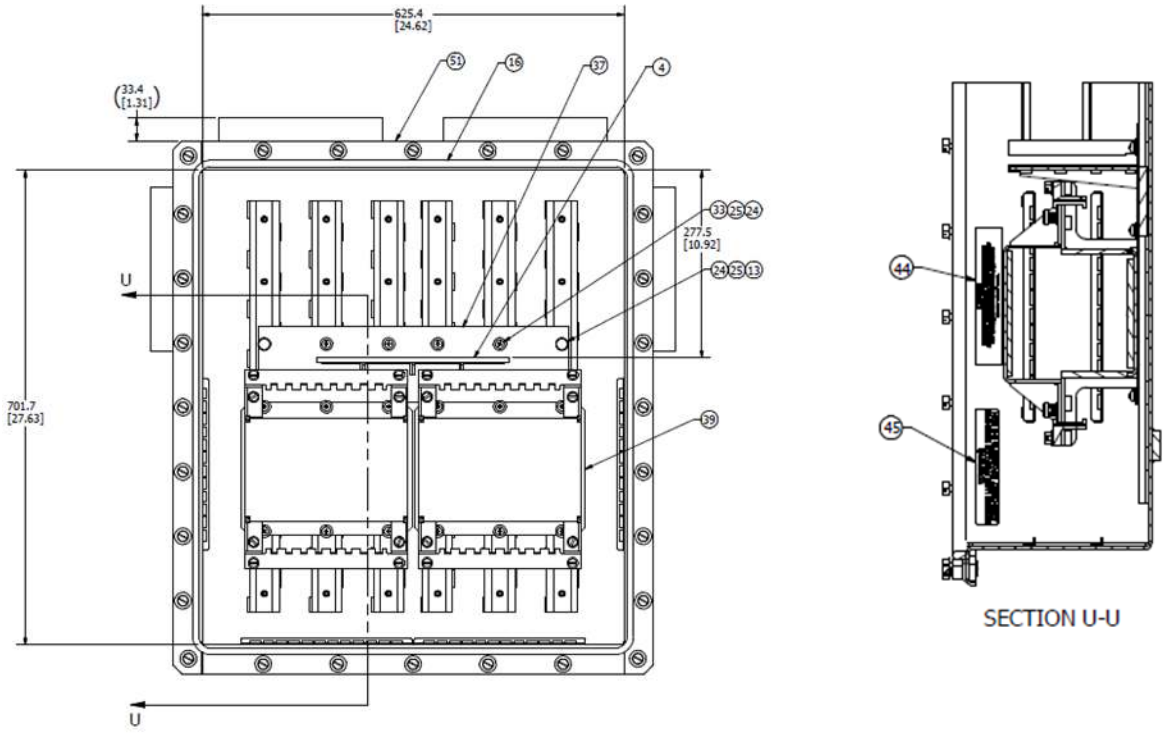
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NOTE: Splice tray holders and splice trays are not included with this part or identifying number (PIN).

FIGURE 4. Enclosure assembly, cover removed, for M24728/12-01.

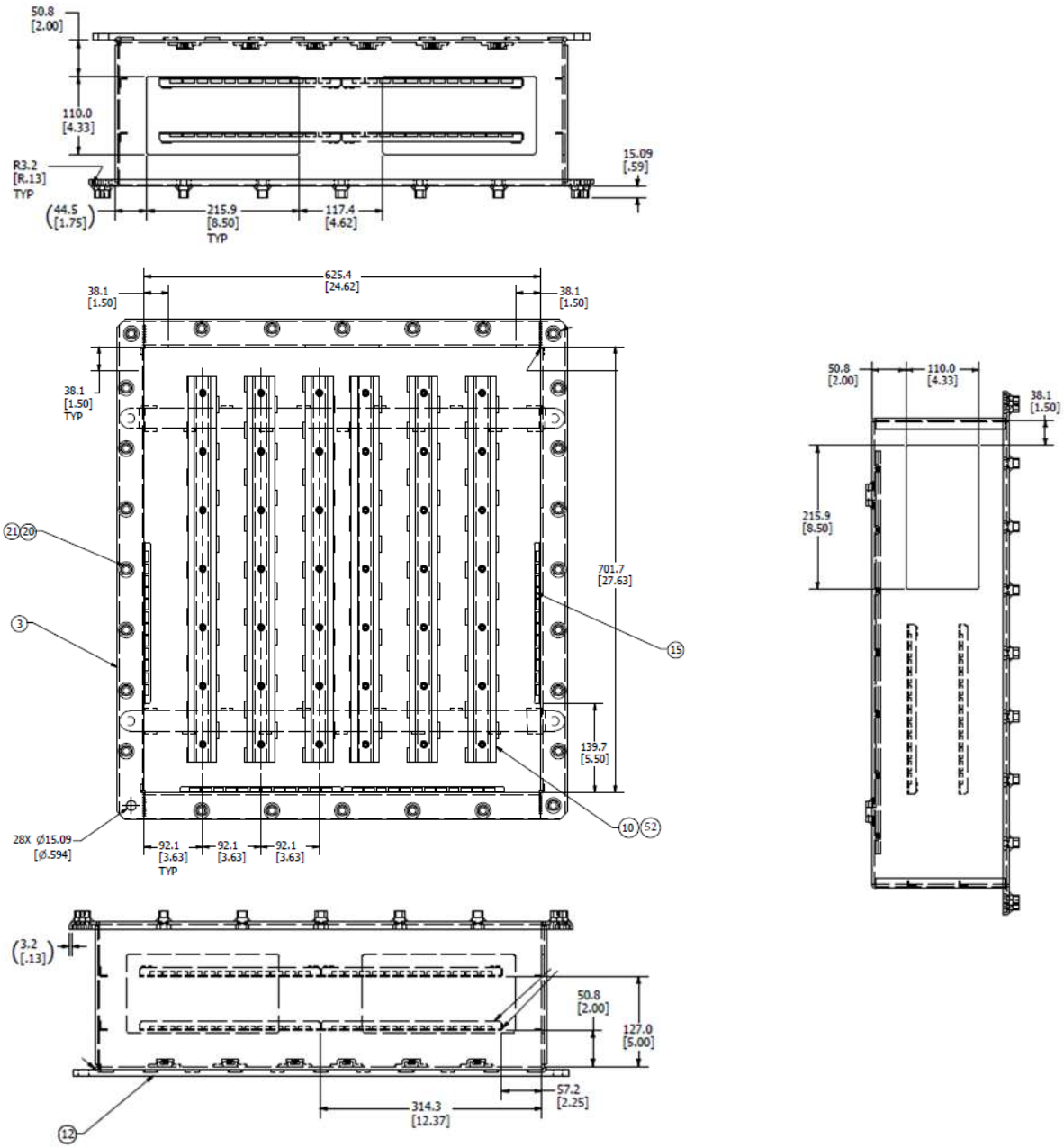
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NOTE: Splice tray holders and splice trays are not included with this PIN.

FIGURE 5. Enclosure assembly, cover removed, for M24728/12-02.

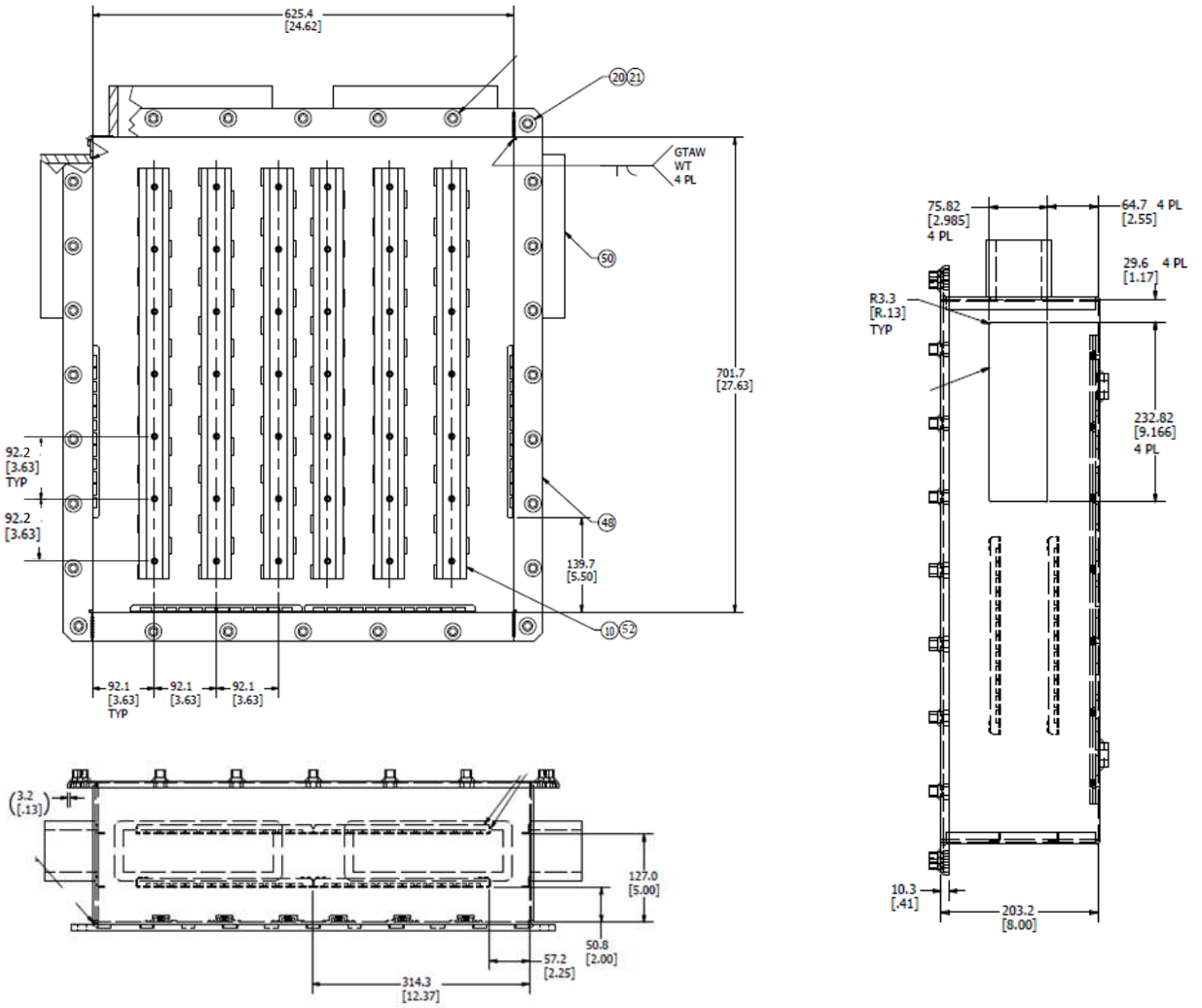
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NOTE: Splice tray holders and splice trays are not included with this PIN.

FIGURE 6. Enclosure assembly, side and end views, for M24728/12-01.

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NOTE: Splice tray holders and splice trays are not included with this PIN.

FIGURE 7. Enclosure assembly, side and end views, for M24728/12-02.

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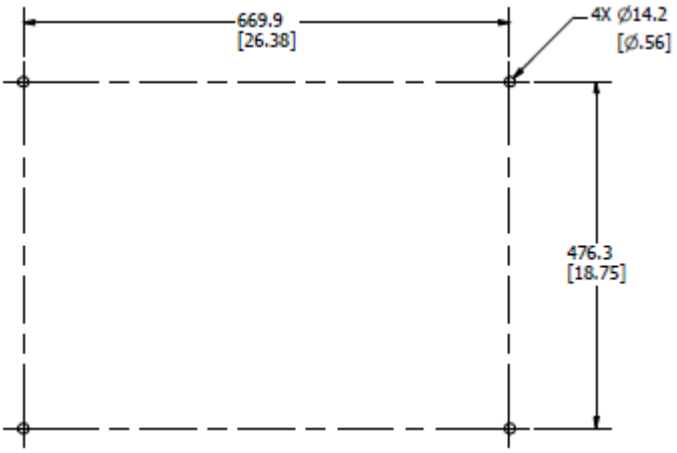


FIGURE 8. Mounting pattern (centered on figure 2).

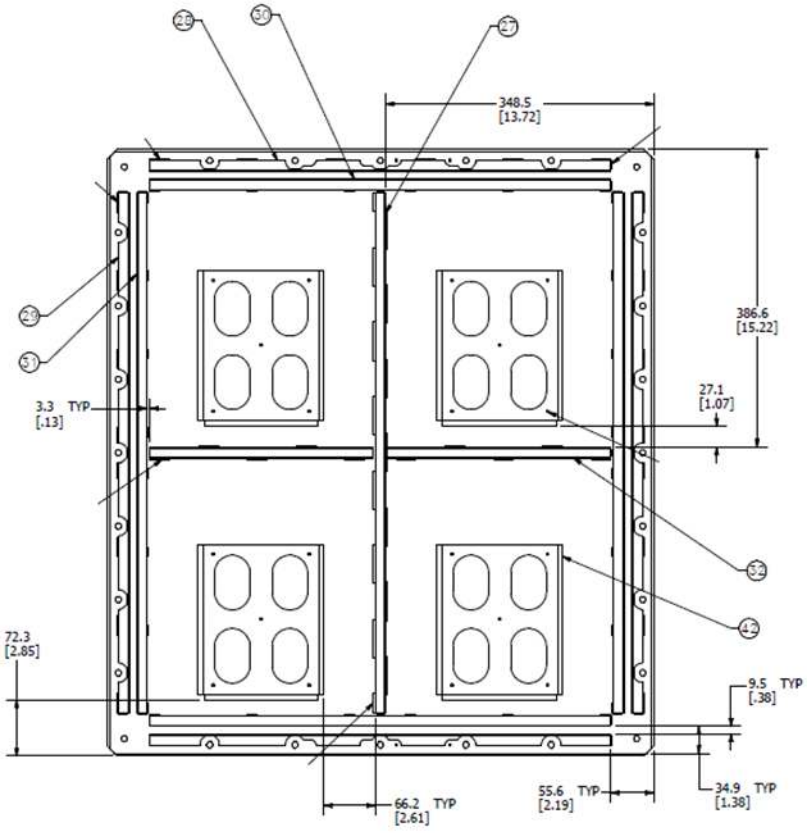


FIGURE 9. Fiber optic diagram holders and cover.

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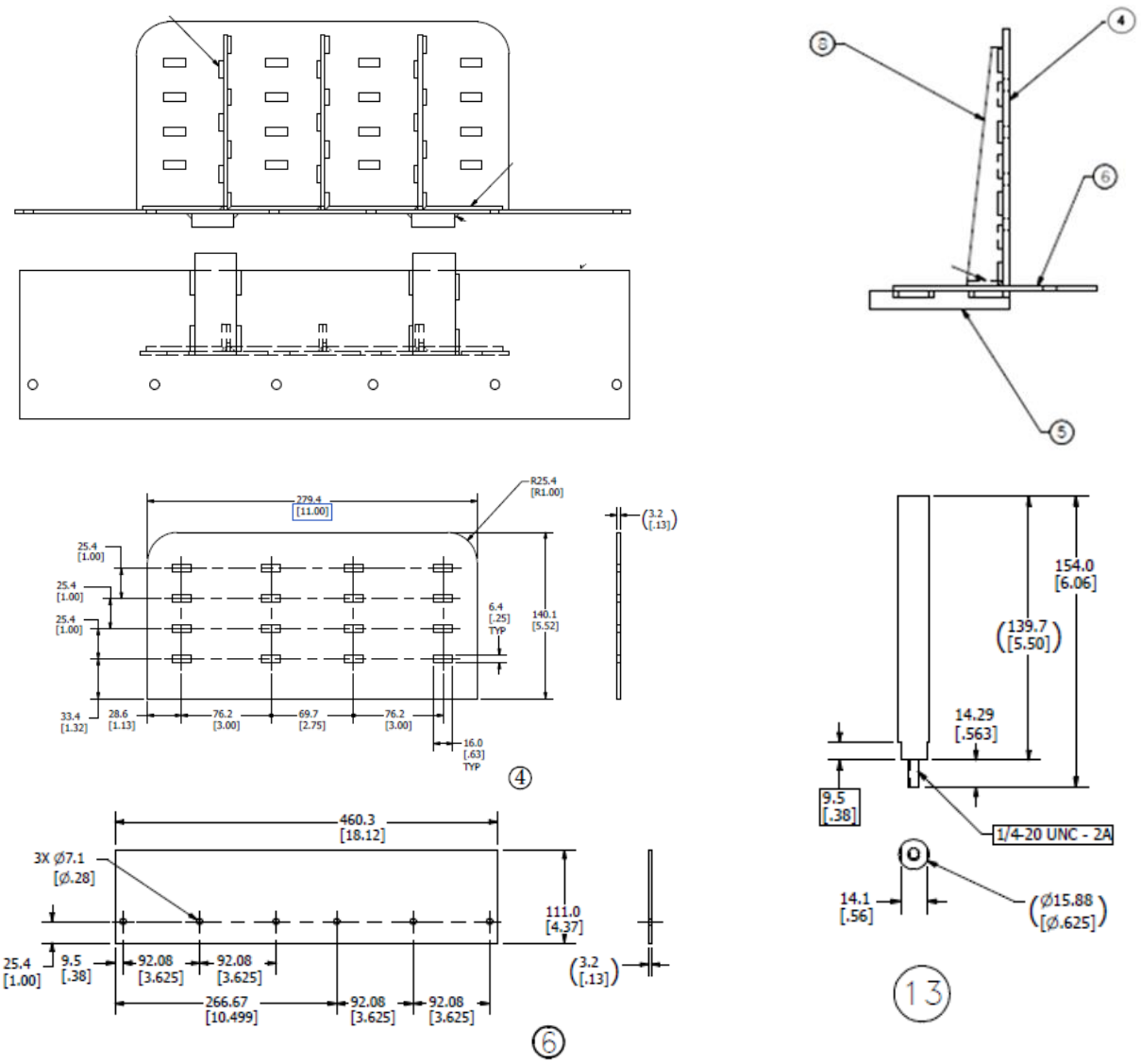


FIGURE 10. Cable management pieces.

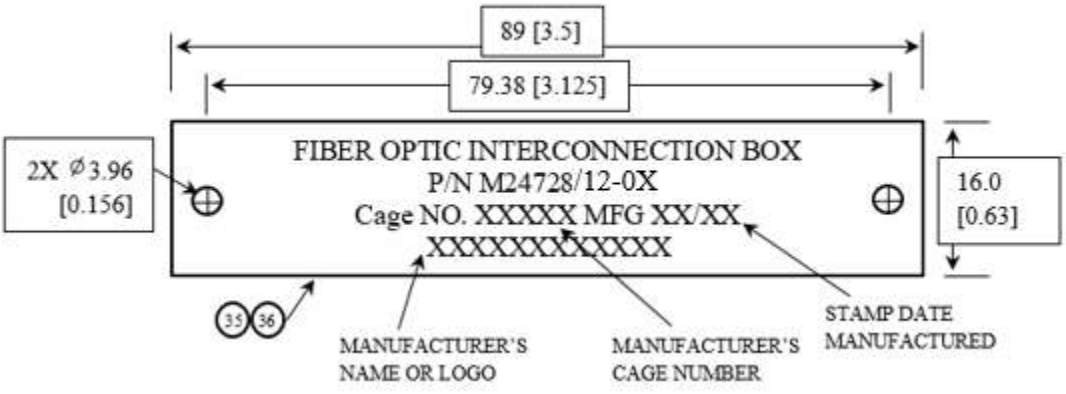


FIGURE 11. Manufacturer's nameplate.

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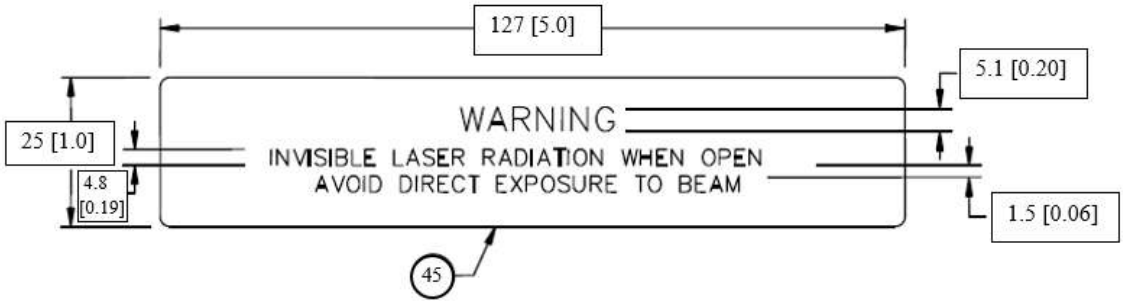


FIGURE 12. Inside warning label.

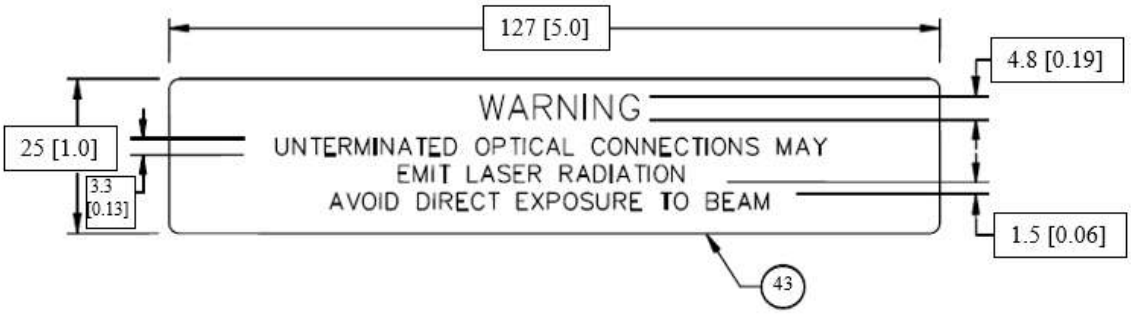


FIGURE 13. Outside warning label.

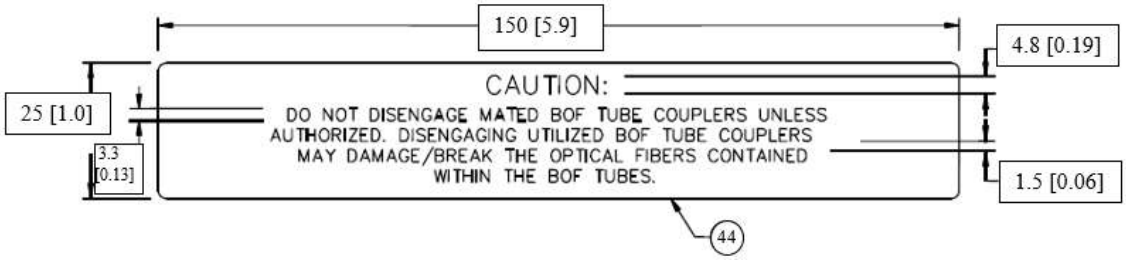


FIGURE 14. Inside and outside caution label.

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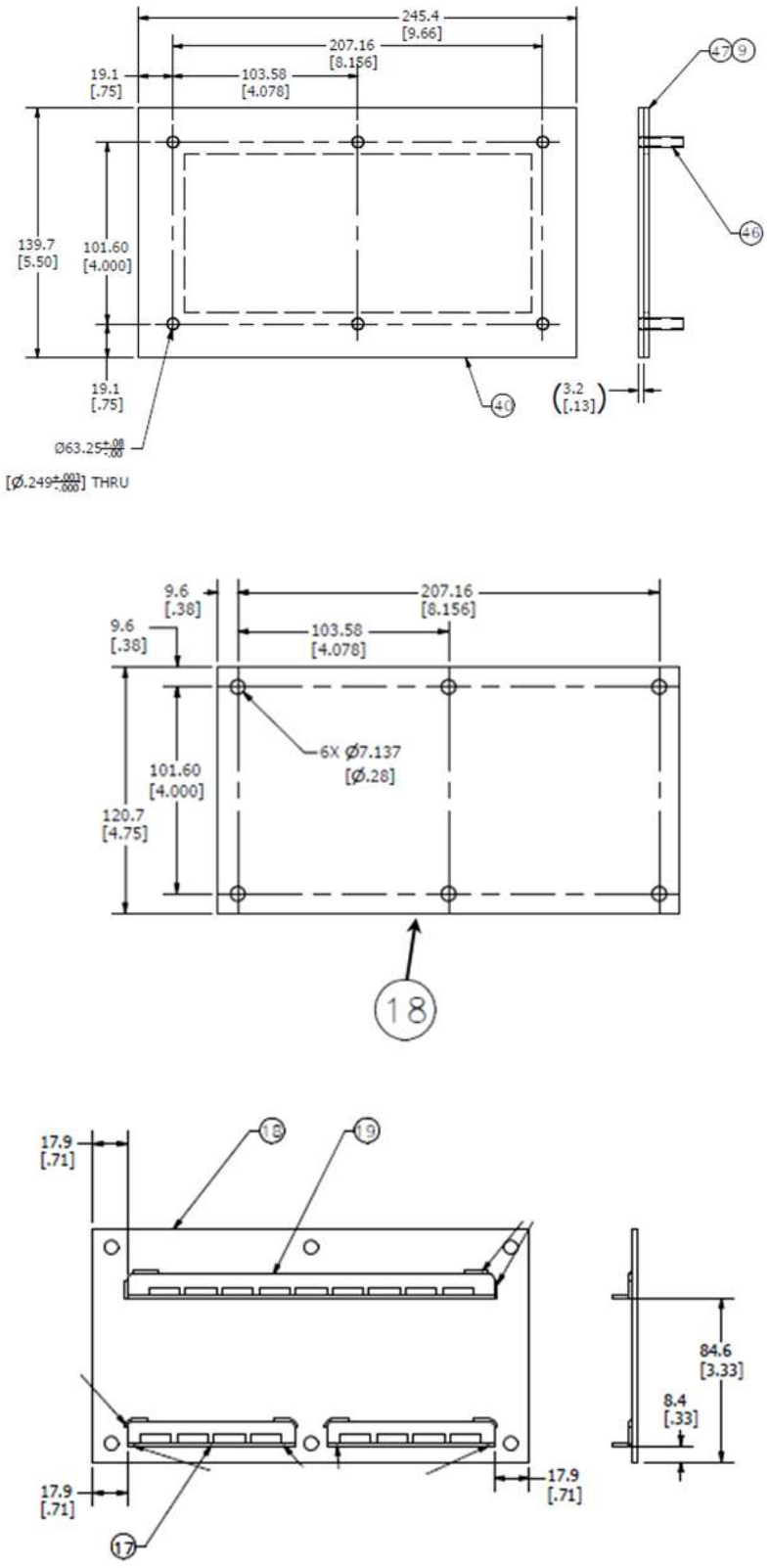


FIGURE 15. Multiple cable transit (MCT) closeout plate assembly and backup plate (M24728/12-01 only).

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TABLE I. Parts list. [1/]

Subcomponent part identification number (circled, figures 1 through 15)	Subcomponent part name	Located in figure(s)	Quantity	Material	Remarks
1	Enclosure assembly (M24728/12-01)	4	1	See remarks	Comprised of the fiber interconnection box assembly without the cover
2	Cover	2 and 3	1	Aluminum sheet, 5052-H32	3.2 mm (0.125 in) thick; ASTM B209
3	Box tack-up	6	1	Aluminum sheet, 5052-H32	ASTM B209
4	Face, bulkhead	4, 5, and 10	1	DTI SL6-0100-ER0201, or equal	3.2 mm (0.125 in) thick; P/O [2/]subcomponent 37; ASTM B209
5	Brace, bulkhead	10	1	Aluminum bar, 6061-T6	9.525 x 31.75 mm (0.375 x 1.25 in); P/O subcomponent 37; ASTM B221
6	Base, bulkhead	10	1	Aluminum sheet, 5052-H32	3.2 mm (0.125 in) thick; P/O subcomponent 37; ASTM B209
7	Intentionally left blank	---	---	---	---
8	Gusset, bulkhead	10	1	Aluminum sheet, 5052-H32	3.2 mm (0.125 in) thick P/O subcomponent 37; ASTM B209
9	Adhesive	15	AR	Loctite 404 Quick Set, or equal	---
10	Mounting channel, long	6 and 7	6	Aluminum sheet, 5052-H32	3.2 mm (0.125 in) thick; ASTM B209
11	Close out plate assembly	4	4	See remarks	Comprised of subcomponents 40, 46, and 47
12	Mounting bar	6	2	Aluminum bar, 6061-T6	9.525 x 31.75 mm (0.375 x 1.25 in); ASTM B221
13	Support post	4, 5, and 10	2	CRES rod, type 316	15.875 mm (⅝ in) diameter; ASTM A276/A276M
15	Tie wrap bracket	6	8	Aluminum, 5052-H32	ASTM B209
16	Gasket	4 and 5	1	Extruded rubber	Durometer 65±5; MIL-PRF-6855, class 1; MIL-E-24142
17	Tie wrap bracket, short	15	2	Aluminum sheet, 5052-H32	1.6 mm (0.063 in) thick; quantity per item: 38; ASTM B209

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TABLE I. Parts list – Continued.

Subcomponent part identification number (circled, figures 1 through 15)	Subcomponent part name	Located in figure(s)	Quantity	Material	Remarks
18	Backup plate	15	1	Aluminum sheet, 5052-H32	3.2 mm (0.125 in) thick; quantity per item: 38; ASTM B209
19	Tie wrap bracket, long	15	1	Aluminum sheet, 5052-H32	1.6 mm (0.063 in) thick; quantity per item: 38; ASTM B209
20	Flange pad	6 and 7	28	Aluminum, 5052-H32	ASTM B209
21	Helicoil	6 and 7	28	CRES	⅜-16 x 0.38 in insert
22	Hex bolt	2 and 3	28	CRES	⅜-16 x 1.25 in, slotted
23	Flat washer	2 and 3	28	CRES	9.525 mm (⅜ in)
24	Flat washer	4 and 5	30	CRES	6.350 mm (¼ in)
25	Lock washer	4 and 5	30	CRES	6.35 mm (¼ in) split ring
26	Hex nut	4	24	CRES	¼-20 in
27	Stiffener, long	9	1	Aluminum sheet, 5052-H32	1.6 mm (0.063 in) thick; ASTM B209
28	Gasket angle, short, notched	9	2	Aluminum sheet, 5052-H32	1.6 mm (0.063 in) thick; ASTM B209
29	Gasket angle, long, notched	9	2	Aluminum sheet, 5052-H32	1.6 mm (0.063 in) thick; ASTM B209
30	Gasket angle, short	9	2	Aluminum sheet, 5052-H32	1.6 mm (0.063 in) thick; ASTM B209
31	Gasket angle, long	9	2	Aluminum sheet, 5052-H32	1.6 mm (0.063 in) thick; ASTM B209
32	Stiffener, short	9	2	Aluminum sheet, 5052-H32	1.6 mm (0.063 in) thick; ASTM B209
33	Screw, pan head Phillips	4 and 5	4	CRES	¼-20 x 0.63 in
34	Screw, pan head Phillips	2 and 3	4	CRES	6-32 x 0.25 in
35	Blank nameplate	2, 3, and 11	1	Aluminum IAW 3/ SAE AMS-QQ-A-250/8	Nameplate made IAW MIL-DTL-15024
36	ID plate	2, 3, and 11	1	Aluminum IAW SAE AMS-QQ-A-250/8	ID plate made IAW MIL-DTL-15024
37	Bulkhead assembly	4 and 5	1	See remarks	Comprised of subcomponents 4, 5, 6, and 8

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TABLE I. Parts list – Continued.

Subcomponent part identification number (circled, figures 1 through 15)	Subcomponent part name	Located in figure(s)	Quantity	Material	Remarks
38	Backup plate assembly	4	4	See remarks	Comprised of subcomponents 17, 18, and 19
39	Splice tray and holder	4 and 5	N/A	See remarks	Shown for reference. Not part of any PIN procurement.
40	Closeout plate	15	4	Aluminum sheet, 5052-H32	3.2 mm (0.125 in) thick; ASTM B209
42	Diagram holder	9	4	Aluminum sheet, 5052-H32	1.6 mm (0.063 in) thick; ASTM B209
43	Warning notice label, outside	2, 3, and 13	1	Adhesive-backed metal foil	MIL-DTL-15024, type G
44	Caution notice label	2, 3, 4, 5, and 14	2	Adhesive-backed metal foil	MIL-DTL-15024, type G
45	Warning notice label, inside	4, 5, and 12	1	Adhesive-backed metal foil	MIL-DTL-15024, type G
46	Clinch stud	15	6	CRES	¼-20 UNC-2A x 1.00LG
47	Gasket	15	1	Rubber sheet	1.6 mm (0.063 inch) thick; durometer 60; MIL-PRF-6855, class 1, grade 60; MIL-PRF-6855/1
48	Box tack-up	7	1	Aluminum sheet, 5052-H32	3.2 mm (0.125 in) thick; ASTM B209
50	180-mm mini-transit frame	7	4	Aluminum, 5052-H32	Welded frame; ASTM B209; see MIL-DTL-24705/1 for guidance
51	Box weldment	5	1	Aluminum sheet, 5052-H32	ASTM B209
52	Channel and clinch nut assembly	6 and 7	6	Aluminum sheet, 5052-H32	ASTM B209

NOTES:

1/ All parts called out in table I are applicable to both M24728/12-01 and M24728/12-02, except for subcomponents 11, 38, and 50. Subcomponents 11 and 38 are applicable only to M24728/12-01, whereas subcomponent 50 is applicable only to M24728/12-02.

2/ Part of (P/O).

3/ In accordance with (IAW).

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Scope. The detail requirements specified herein cover 711- by 787-millimeter spraytight and submersible (as noted) fiber optic cable routing interconnection boxes suitable for military use.

Part or identifying number (PIN). The PIN consists of the letter "M" followed by the base specification number, the specification sheet number, and a PIN code (see table II).

M	**24728**	**/12**	**-**	**XX**
Prefix for Military Specification	Specification Number	Spec Sheet No.		PIN code (see table II)

TABLE II. PIN code.

PIN code	Description	Figure
01	Spraytight fiber optic interconnection box enclosure assembly, cover, and four removable closeout plate assemblies with mounting hardware.	1, 2, 4, 6, and 8 through 15
02	Submersible fiber interconnection box enclosure assembly, cover, and four welded cable entrances and four wedge packs.	1, 3, 5, and 7 through 14

REQUIREMENTS:

Materials. The enclosure assembly shall be constructed using the materials specified in table I.

Finish. The enclosure assembly shall be coated with a chemical film in accordance with MIL-DTL-5541, type II, class 1A, prior to installing the helicoils in the fiber optic interconnection box assembly. The enclosure assembly shall have one continuous film finish coat of polyester triglycidyl isocyanurate (TGIC) powder coating in accordance with MIL-PRF-24712, type IV, per the guidance of Powder Coating Institute (PCI) Technical Brief #1. The powder coat shall be gray in accordance with SAE-AMS-STD-595, color 26307. The powder coat shall have a minimum film thickness of 2 mils after curing. A finish coat shall be applied prior to installing all gaskets, labels, and nameplates.

Nonmagnetic materials. Not applicable.

Design and construction. The design and construction of the interconnection box shall be in accordance with table I and as shown on figures 1 through 15.

Fiber optic interconnection box mounting. The fiber optic interconnection box shall be mounted as shown on figures 4 through 7.

Cable interconnection interface.

Submersible fiber optic interconnection boxes with MCT. The cable feed-through shall be in accordance with MIL-DTL-24705 and shall accept cables having outer diameters between 6 millimeters (0.24 inch) and 32.5 millimeters (1.28 inches) constructed in accordance with MIL-PRF-85045 for submersible fiber optic interconnection boxes. Insert blocks compatible with the MCT frame are specified in MIL-DTL-24705/1 for submersible fiber optic interconnection boxes.

Spraytight fiber optic interconnection boxes. Cable entry, whether through a removable MCT in accordance with MIL-DTL-24705/4 or nylon stuffing tubes in accordance with ASTM F1836M, shall accept cables having outer diameters between 7.75 millimeters (0.31 inch) and 32.5 millimeters (1.28 inches) constructed in accordance with MIL-PRF-85045. Insert blocks compatible with the MCT frame are specified in MIL-DTL-24705/4 for spraytight fiber optic interconnection boxes.

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Interconnect organization.

Fiber optic diagram holders. Fiber optic diagram holders shall be as shown on figure 9.

Fiber optic interconnection modules. The fiber optic interconnection box shall provide sufficient space for accommodating the items listed in table III.

Mass. The mass of the interconnect organization shall be not greater than 45 kilograms (100 pounds).

Stress relief. Enclosure assemblies shall be stress-relieved in accordance with MIL-E-24142.

TABLE III. Fiber optic interconnection modules table.

	M24728/8-50 1/	**M24728/11-01 1/**	**M24728/13-01 2/**
Number of splice trays (max splices per fiber optic interconnection box)	8 (96)	16 (192)	16 (384)
Max number of splice tray holders	2	2	2
Max number of connector modules	N/A	N/A	N/A
Number of fiber optic connector adapters	N/A	N/A	N/A

NOTES:

1/ Assuming use of M24728/8-51 splice tray with 12 splices per tray.

2/ Assuming use of M24728/13-02 splice tray with 24 splices per tray.

Cable organization. Cable organization shall be as shown in table IV.

TABLE IV. Cable organization.

Tie wrap rails	Tie wrap rails shall be as shown on figures 4, 5, 6, 7, and 15.
Tube routing region	The tube routing regions shall provide for routing of MIL-PRF-85045 cables, including blown optical fiber (BOF) tube cable. The tube routing region of the fiber optic interconnection box shall limit BOF tube bend diameter to 127 millimeters (5 inches) for 8 millimeters tubing and 102 millimeters (4 inches) for 5 millimeters tubing.
Interconnect region	Limit the BOF tube bend diameter to 102 millimeters (4 inches) for 5 millimeters tubing. The interconnect region shall also limit the bend diameter of optical fiber cable components (OFCC) in accordance with MIL-PRF-85045/16.
Cable management	Cable management shall be accomplished using bulkheads, routing posts, and tie wrap rails (subcomponents 5, 9, and 11, respectively); see figures 2 and 3 for locations.

Change in optical transmittance. Applicable with the following modification: The change in optical transmittance of splices installed in the fiber optic interconnection box in a standard manner shall be not greater than the maximum specified value of the change in optical transmittance for fusion splices that are in accordance with MIL-PRF-24623/6.

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Cable retention. Not applicable.

Cable seal flexing. Not applicable.

Cable twist. Not applicable.

Compression resistance. Not applicable.

Impact resistance. Not applicable.

Salt spray. Not applicable.

Operating and storage temperature range. The operating temperature range shall be -28 to 65 °C and the storage temperature range shall be -40 to 70 °C.

Structural integrity. Applicable with the following modification: The assembled fiber optic interconnection box shall have a resonant frequency not less than the maximum frequency specified when tested in accordance with MIL-STD-167-1, type I.

Flame spread. Not applicable.

Flammability. Not applicable.

Qualification item configuration. When configuring the fiber optic interconnection box for qualification, it shall have two removable MCTs in accordance with MIL-DTL-24705/4. A minimum of two M24728/13-01 splice tray holders shall be installed and populated with M24728/13-02 splice trays. Optical fiber splices assembled on M85045/16-02 cables in accordance with MIL-PRF-24623/6 shall be stored in the splice trays being optically monitored.

VERIFICATION:

Qualification inspection and sample size. Fiber optic interconnection boxes shall be inspected for qualification as specified in table V.

Conformance inspection. Conformance inspection shall consist of the inspections and tests specified for groups A, B, and C (table V).

TABLE V. Qualification and conformance inspections.

Inspection	Requirement paragraph in MIL-DTL-24728	Verification method paragraph in MIL-DTL-24728	Qualification sample size	Conformance sample size	Conformance inspection group [5/]
Group I					
Visual and mechanical	3.4, 3.7	4.6.1	1	100% [4/]	A
Structural integrity	3.5.3.7.1	4.6.1	1	-	-
Mass	3.4.8	4.6.3.1	1	100% [4/]	A
Size	3.4.9	4.6.3.2	1	100% [4/]	A
Identification markings	3.6	4.6.1	1	100% [4/]	A
Group II					
Temperature life	3.5.3.1	4.6.6.1	1	1	C

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TABLE V. Qualification and conformance inspections – Continued.

Inspection	Requirement paragraph in MIL-DTL-24728	Verification method paragraph in MIL-DTL-24728	Qualification sample size	Conformance sample size	Conformance inspection group 5/
Thermal shock	3.5.3.2	4.6.6.2	1	1	C
Operating temperature	3.5.3.10	4.6.6.10	1	1	C
Temperature/ humidity cycling	3.5.3.3	4.6.6.3	1	1	C
Fluid immersion	3.5.3.6	4.6.6.6	1 1/	1	C
Shock	3.5.3.8	4.6.6.8	1	1 3/	C
Vibration	3.5.3.7	4.6.6.7	1	1 3/	C
Water pressure	3.5.3.5	4.6.6.5	1	1 2/	B
Fungus resistance	3.5.4.1	4.6.7.1	1 1/	-	-

NOTES:
1/ Fiber optic interconnection box material samples may be used for this test.
2/ Compliant group B conformance tested samples may still be delivered.
3/ Shock and vibration conformance tests shall have one M24728/13-01 module and two M24728/13-02 splice trays installed for monitoring of optical performance.
4/ Unless alternative sampling plan has been approved (MIL-DTL-24728).
5/ Qualification by similarity considerations may be applied to conformance inspection for group C tests.

Qualification by similarity. Applicable. Manufacturers who produce product for MIL-DTL-24728/12, who are qualified under M24728/12-01, and pass shock and water tight integrity testing and group I inspections as specified within for the M24728/12-02 are qualified under this specification for M24728/12-02. This qualification by similarity is applicable if the same materials and construction are used in the previously qualified M24728/12-01 box and the box under test.

Manufacturers who produce product for MIL-DTL-24728/12, who are qualified under M24728/12-01 and M24728/3, and pass water tight integrity testing and group I inspections as specified within for the M24728/12-02 are qualified under this specification for M24728/12-02. This qualification by similarity is applicable if the same materials and construction are used in the previously qualified M24728/12-01 and M24728/3 boxes and the box under test.

Change in optical transmittance. Applicable with the following modification: The change in optical transmittance shall be tested in accordance with MIL-STD-1678-2, measurement 2102.

Discontinuity. No discontinuity shall occur when the fiber optic interconnection box is tested in accordance with MIL-STD-1678-2, measurement 2104, using equipment having a time resolution sufficient to resolve discontinuities of duration not less than 50 microseconds. A discontinuity is a reduction of signal strength for the duration specified for fusion splices in accordance with MIL-PRF-24623/6.

Temperature life. Applicable with the following modifications:

a. Testing shall be performed in accordance with TIA/EIA-455-5, method B. The sub-cycle shall be included.

b. The specimen shall be exposed to dry air at 85 +3/-0 °C (185 +5/-0 °F) for a period of 240 hours.

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c. Post-test dimensional inspection shall be limited to the height, length, and width of the fiber optic interconnection boxes (see figures 2 through 5), and the length and width of the cover (see figures 2 through 5).

d. The requirements for change in optical transmittance are not applicable.

Thermal shock. Applicable with the following modifications:

a. Testing shall be performed in accordance with TIA-455-71, test condition C-0.

b. Post-test dimensional inspection shall be limited to the height, length, and width of the fiber optic interconnection boxes (see figures 2 through 5), and the length and width of the cover (see figures 2 through 5).

c. The requirements for change in optical transmittance are not applicable.

Temperature/humidity cycling. Applicable with the following modification: The requirements for change in optical transmittance are not applicable.

Water pressure.

Spraytight (applicable to M24728/12-01 only). The enclosure shall pass the spraytight test in accordance with MIL-STD-108.

Submersible. Applicable to the submersible version, M24728/12-02. The applied pressure shall be 44.8 kilopascals (6.5 pounds per square inch), equivalent to a depth of 4.57 meters (15 feet). The interconnection cover bolts and the MCT wedge pack nut shall be tightened to a minimum torque of 16.94 newton-meters (150 inch-pounds) during this test.

Vibration. Applicable with the following modification: The vibration test shall be performed with optical fiber splices assembled on M85045/16-02 cables in accordance with MIL-PRF-24623/6 that are stored in the splice tray and splice tray holder. The fiber optic interconnection box shall meet the discontinuity requirements during the test and the change in optical transmittance requirements after the test.

Shock. Applicable with the following modification: Optical fiber splices shall meet the discontinuity requirements during the test and the change in optical transmittance requirements after the test.

Operating temperature. Applicable with the following modifications:

a. Testing shall be performed in accordance with TIA-455-3 using the test condition schedule and soak times in table VI and at the operating temperature range specified herein.

b. Optical fiber splices shall meet the change in optical transmittance requirements during and after the test.

c. A post-test visual examination of the test specimens shall reveal no leakage of waterproofing compounds or other loss of sealing capability, no surface or identification marking impairment, and no damage detrimental to the operation of the test specimens.

TABLE VI. Operating temperature schedule.

Step	Action	Temperature	Duration
1	Maintain	Room ambient	4 hours (minimum)
2	Ramp to	Low operating temp +0/-3 °C (+0/-5 °F)	2 hours
3	Maintain	Low operating temp +0/-3 °C (+0/-5 °F)	8 hours (minimum)
4	Ramp to	25±3 °C (77±5 °F)	2 hours
5	Maintain	25±3 °C (77±5 °F)	8 hours (minimum)
6	Ramp to	High operating temp +0/-3 °C (+0/-5 °F)	1 hour
7	Maintain	High operating temp +0/-3 °C (+0/-5 °F)	8 hours (minimum)

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TABLE VI. Operating temperature schedule – Continued.

Step	Action	Temperature	Duration
8	Ramp to	25±3 °C (77±5 °F)	1 hour
9	Maintain	25±3 °C (77±5 °F)	8 hours (minimum)
10	Repeat steps 2 through 9 four additional times for a total of five cycles.		

INTENDED USE. The cable-routing fiber optic interconnection boxes are designed with the tube routing region to accommodate a maximum of eight MIL-PRF-85045/25 BOF tube cables. The tube routing region of the fiber optic interconnection boxes should be used for routing and interconnecting BOF tube cable. The fiber optic interconnection boxes also contain an interconnect region intended for the routing of OFCCs and 5-millimeter BOF tubing.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85

Review activities:
Army – AR, MI
Navy – AS, CG, MC, YD
Air Force – 19
DLA – CC, GS

Civil agency:
NASA – NA

Preparing activity:
Navy – SH
(Project 6099-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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